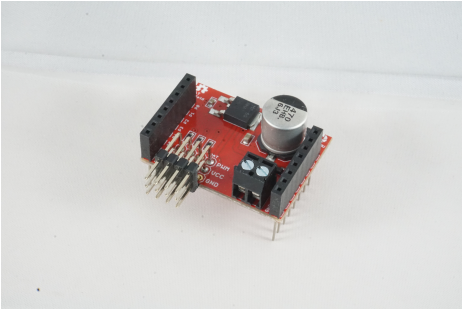
OpenMV® Pan and Tilt Shield

Datasheet

SKU: OMV-SHIELD-PANDT-V1

Description



The Pan and Tilt Shield gives your OpenMV Cam the ability to control three servo motors (or ESCs) directly along with providing flexible power options. You can power your OpenMV Cam via an unregulated battery (regulated down to 5 V) while also providing 5 V to all four servo headers, or power via the servo headers from an ESC supplying 5\u20136 V. If you are building a pan and tilt setup or making a DIY robocar to race, this shield is all you need to get everything wired up.

Target Areas

Pan-tilt systems, gimbal control, RC vehicles, robocars, DIY racing, camera positioning, servo-driven mechanisms.

Key Features

- **3 servo outputs** — Control three servo motors or ESCs directly.
- **Flexible power** — Battery-regulated 5 V or ESC-powered 5–6 V options. All four servo headers powered.
- **Compatible with Servo Shield** — Can be used simultaneously with the Servo Shield (different pins).

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1 Features

1.1 Servo Control

Component	Details
Outputs	3 servo motor or ESC outputs with 4 servo headers.
Power Options	Option 1: Unregulated battery regulated to 5 V for camera and servos. Option 2: ESC providing 5–6 V to camera and servos via servo headers.

1.2 Compatibility

Component	Details
Supported Cameras	All OpenMV Cam models.
Shield Stacking	Compatible with Servo Shield (different pins). NOT compatible with Motor Shield.

2 General Info

Dimension	Value
Weight	30 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/pan-and-tilt-shield
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.